Quantifying Voice Impairment in Parkinson's Disease: Acoustic Evidence from Sustained Vowel Production

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Parkinson's Disease frequently leads to hypokinetic dysarthria, a speech disorder characterized by dysphonia and reduced phonatory control. Acoustic analysis provides a window into the laryngeal impairment accompanying this pathology. This study examined phonatory impairment in sustained vowel /a/ production using a publicly available dataset of individuals with Parkinson's disease (PwPD; N = 40) and healthy controls (HC; N = 41). Acoustic features were extracted in Praat and then analyzed in R. A MANOVA revealed significant multivariate group differences across the acoustic feature set (Pillai's Trace = 0.22, p = .03). However, no individual measure survived Holm correction, although mean F0 showed a marginal raw difference (p = .02; adjusted p = .15). These findings suggest that while PwPD and HC differ subtly across a combined acoustic profile, isolated features such as jitter, shimmer, and harmonicity may not reliably distinguish the groups in sustained phonation tasks. Results highlight the value of multivariate approaches to voice analysis and underscore the need for larger datasets, age-matched samples, and more ecologically valid speech tasks to characterize dysphonia in PD.

Keywords: vocal stability, f0, amplitude, harmonicity, Parkinson's, vowels

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Parkinson's disease pathology

Parkinson's disease (PD) is a chronic, progressive neurodegenerative disorder primarily characterized by the loss of dopaminergic neurons (Ramesh & Arachchige, 2023). Dorsey et al. (2018) expect that by the year 2040, approximately 12.9 million individuals will be diagnosed with PD. It affects millions worldwide, with prevalence among older populations (Li et al., 2025). People with Parkinson's disease (PwPD) often experience impairments across multiple spheres. Non-motor symptoms include cognitive decline, sleep disorders, autonomic dysfunction, and mood disturbances (Poewe, 2008). Due to the impact on the core motor system, PwPD often experience bradykinesia, resting tremor, rigidity, and postural instability (Ramesh & Arachchige, 2023; Moya-Galé & Levy, 2019; Mamelak, 2018).

Hypokinetic dysarthria

Communication is also frequently impacted. As PD leads to reduced respiration, impaired laryngeal control, rigidity of the vocal folds, and decreased articulatory precision, PwPD can develop communication and speech disorders (Wright & Aharonson, 2025; Moya-Galé & Levy, 2019). Up to 90% of PwPD develop hypokinetic dysarthria, a speech disorder affecting voice and prosody (Convey et al., 2024). It is often characterized by dysphonia, which consists of impaired vocal quality, reduced loudness, breathiness, roughness, monopitch, and instability (Wright & Aharonson, 2025; Convey et al., 2024; Moya-Galé & Levy, 2019).

Acoustic features

Voice production relies on fine motor control, meaning that the neuromotor deficits of PD can be observed in acoustic signals. Studying dysphonia acoustically can provide objective markers of disease severity and communication impairment.

F0 stability

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Fundamental frequency (F0), determined by the rate of vocal fold vibration, is perceptually related to vocal pitch. In PD, reduced voluntary prosodic control produces measurable changes in F0 variability, which is the ability to intentionally modulate pitch over time (Bowen et al., 2013). Rather than examining F0 range alone (which is susceptible to outlier influence), clinically relevant measures of F0 stability include standard deviation of F0 (F0SD), semitone standard deviation (STSD), and frequency perturbation measures such as jitter, relative average perturbation (RAP), and pitch perturbation quotient (PPQ).

The mechanisms underlying reduced F0 variability in PD are multifactorial. Increased stiffness and rigidity of the laryngeal muscles and vocal folds impair the dynamic tension adjustments necessary for pitch modulation. Reduced laryngeal muscle activity and decreased stability of laryngeal structures further constrain the ability to sustain steady phonation. More recent evidence suggests that sensorimotor deficits and internal cueing issues that reflect broader dopaminergic system dysfunction also contribute to these F0 changes (Yang et al., 2020; Bowen et al., 2013). Importantly, dopaminergic medication significantly improves F0 variability, indicating that the loss of dopamine directly mediates this deficit.

Jitter specifically measures cycle-to-cycle frequency variation in successive vocal fold cycles. In PD, jitter values show considerable variation in the literature; some studies report higher jitter reflecting irregular vocal fold oscillation (Chiaramonte & Bonfiglio, 2020), while others find lower jitter associated with creakier voice production due to vocal fold stiffness (Yang et al., 2020). Regardless of direction, changes in jitter correlate with disease severity and reflect the irregular vocal fold muscle contractions and loss of motor control characteristic of hypokinetic dysarthria. These perturbation measures represent early biomarkers of PD; notably, case studies show visible F0 changes up to 5 years before clinical PD diagnosis, suggesting

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acoustic analysis could enable earlier disease detection and longitudinal disease monitoring (Harel et al., 2004).

Amplitude stability

Amplitude stability is the consistency of loudness across successive vocal fold cycles. It is measured through shimmer in cycle-to-cycle amplitude variation and amplitude perturbation quotients through APQ3, APQ5, APQ. Increased shimmer and APQ values are characteristic of PD voice and reflect incomplete or irregular vocal fold closure (Abola & Wolden, 2025). Physiologically, shimmer arises from asymmetrical or variable vocal fold motion during phonation, which generates excessive airflow and produces amplitude fluctuations perceptible as tremor or instability in the voice.

In PD, shimmer is significantly elevated compared to healthy controls. The acoustic shimmer parameters—Shimmer in %, Shimmer in dB, APQ3, APQ5—show normative ranges of ≤3.5%, ≤0.35 dB, ≤3.5%, and ≤4.5% respectively, with PD patients typically exceeding these thresholds (Abola & Wolden, 2025; Abraham & Geetha, 2023). Both frequency perturbation measures (jitter) and amplitude perturbation measures (shimmer) are sensitive enough to distinguish voice problems in PD from healthy speech and correlate with perceptual severity ratings (Abraham & Geetha, 2023).

Harmonicity

In PD, harmonic-to-noise ratio (HNR) is typically lower compared to age-matched controls, though some studies report mixed findings. The physiological basis involves the inadequate or excessive closing of the vocal folds and irregular or asymmetrical vocal fold motion during phonation, resulting in added noise in the acoustic signal. This noise generation reflects the

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combination of increased laryngeal rigidity and the loss of precise motor control over vocal fold contact patterns.

The variability in findings may reflect methodological differences in HNR calculation and heterogeneity in PD presentation, such as medication status, disease severity, presence of overt dysarthria (Cao et al., 2025; Wright & Aharonson, 2025; Convey et al., 2024; Yang et al., 2020; Moya-Galé & Levy, 2019). Despite inconsistencies, HNR remains clinically valuable as one of several acoustic features sensitive to dysphonia and correlating with disease severity.

Duration

Temporal characteristics of speech are substantially altered in PD and reflect the bradykinesia and rigidity affecting respiratory and laryngeal systems. Speech rate is typically slower in PD compared to controls, accomplished through both reduced articulation rate and increased pause duration (Cao et al., 2025; Wright & Aharonson, 2025; Convey et al., 2024; Yang et al., 2020; Moya-Galé & Levy, 2019). Duration may also be task-dependent when used in cases other than natural speech.

Current study

Despite extensive research, sustained-vowel dysphonia remains analytically valuable because it provides controlled, reliable data to assess phonatory impairment. Given the physiological mechanisms impacted by PD, objective acoustic analysis can help characterize the severity of hypokinetic dysarthria and potentially contribute to clinical monitoring.

This mini project investigates how PwPD differ from healthy norms across key acoustic dimensions—F0 stability, amplitude stability, noise ratios—using sustained vowels to provide clear phonatory markers. It is expected that, in comparison to healthy controls (HC), PwPD will show (1) reduced F0 stability, (2) decreased amplitude stability, and (3) lower harmonicity.

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Methods

Participants

Participants were selected from the open dataset "Voice Samples for Patients with Parkinson's Disease and Healthy Controls" compiled by Prior et al. (2023). Some were classified as PwPD (N = 40), while others were considered healthy controls (HC; N = 41). Age (PwPD = 67; HC = 48) and sex were recorded for each participant.

Procedure

Collection

Voice recordings are in .wav format and represent a prolonged enunciation of the vowel /a/. These recordings were made using participant's telephones. Phone distance from mouth was not controlled.

Analysis

Praat

Acoustic analysis was performed using Praat 6.4.47. Extracted features included mean F0, F0 standard deviation, jitter (Local; Local, absolute; relative average perturbation (RAP); ppq5), shimmer (Local; apq5), harmonic-to-noise ratio (HNR), noise-to-harmonic ratio, and duration. Values were then saved to a CSV file. This process was streamlined through a script (Appendix A).

Statistical analyses

The raw data was then analyzed through R version 4.4.3 with a script (Appendix B). To reduce Type I error rates and account for correlated acoustic variables, MANOVA with Pillai's Trace was conducted. In the case of significance, Welch's t-tests with Holm correction were performed. Alpha values remained at 0.05 across all tests and corrections.

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Results

Descriptive statistics

Data was collected then divided between PwPD and HC (Figure 1).

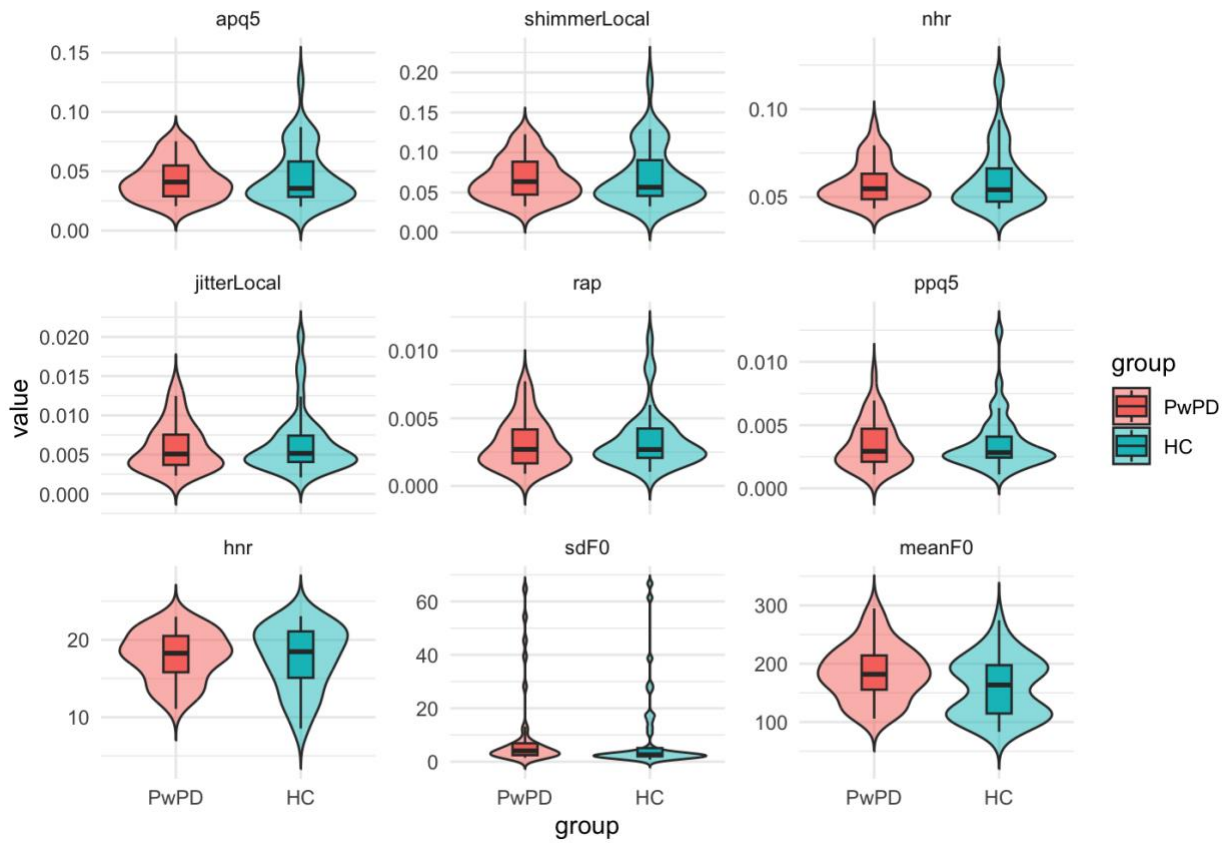

Figure 1. Distribution of raw data for apq5, shimmer (local), NHR, jitter (local), RAP, ppq5, HNR, F0 (standard deviation), and mean F0 but separated by group.

F0 stability

The PwPD yielded an average F0 of 185.05 ± 47.89 Hz, while the HC group yielded 158.63 ± 49.83 Hz. The local jitter average ranged from 0.60 ± 0.31% for PwPD to 0.63 ± 3.82% for HC. Average RAP was 0.32 ± 0.18% for PwPD and 0.33 ± 0.21% for HC. The five-period perturbation quotient (ppq5) were similar between groups (PwPD = 0.35 ± 0.19%; HC = 0.36 ± 0.21%).

Amplitude stability

Local shimmer measures yielded 6.74 ± 2.57% for PwPD and 7.03 ± 3.56% for HC. PwPD had a five-point amplitude perturbation quotient (apq5) of 4.27 ± 1.63%, while HC had 4.60 ± 2.39%.

Harmonicity

PwPD were found to have an average HNR of 17.90 ± 3.19 dB but an average NHR of 0.06 ± 0.01. The HC had an average HNR of 17.63 ± 4.09 dB but an average NHR of 0.06 ± 0.02.

Duration

The PwPD typically had longer sustained vowels with an average of 3.59 ± 1.53 seconds, while the HC group yielded an average time of 3.07 ± 1.07 seconds.

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The MANOVA test detected significant differences between the PwPD and HC based on the aforementioned acoustic features (F = 2.17, p = 0.03, Pillai's Trace = 0.22). This warranted further analysis through t-tests.

Follow-up t-tests

Interestingly, only one individual t-test detected significant differences: the mean F0 (p = 0.02); however, this significance did not withstand Holm correction (adjusted p = 0.15). All other t-tests found no significant differences beginning with raw p-values. (sdF0 = 0.87; jitter, local = 0.69; RAP = 0.69; ppq5 = 0.86; shimmer, local = 0.67; apq5 = 0.47; HNR = 0.73; NHR = 0.40).

Table 1. Follow-up t-tests based on the significance of the MANOVA. Cohen's d and 95% Confidence Interval are provided for each acoustic parameter.

Acoustic Parameter	Raw p-value	Holm p-value	Cohen's d	95% CI
meanF0	0.017	0.151	0.543	[0.097, 0.985]

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sdF0	0.873	1.000	0.036	[−0.400, 0.471]
jitterLocal	0.689	1.000	−0.089	[−0.524, 0.347]
rap	0.694	1.000	−0.087	[−0.523, 0.348]
ppq5	0.860	1.000	−0.039	[−0.474, 0.396]
shimmerLocal	0.668	1.000	−0.095	[−0.525, 0.341]
apq5	0.469	1.000	−0.161	[−0.596, 0.276]
hnr	0.734	1.000	0.076	[−0.360, 0.511]
nhr	0.399	1.000	−0.188	[−0.623, 0.249]

Discussion

MANOVA findings reflect the combined, correlational pattern across these acoustic features. Even though individual variables showed little significant effects, their collective covariance structure differs between groups. This suggests a shared but subtle motor change across laryngeal control mechanisms. Such changes have been recorded in other studies and are worth exploring more in-depth in the following sections.

Acoustic features

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Although mean F0's Holm-corrected $p = 0.15$ technically falls short of the $p < 0.05$ threshold after multiple-comparison correction, its raw p-value of 0.02 remains noteworthy and represents a borderline finding warranting replication and discussion. PD may only affect the pitch-generating mechanism reflected in mean F0 and not cycle-to-cycle stability or amplitude consistency measured in jitter and shimmer, respectively.

Furthermore, no significant difference in jitter and shimmer may indicate a specific disease stage or phenotype of PD. This finding, while contradicting older research (Darley et al., 1969; for a review: Wright & Aharonson, 2025), appears to align with recent findings (Kacha et al., 2017; Rusz et al., 2011; Skodda et al., 2011). A fine-grain examination of PD and its progression as it relates to acoustic features should be examined.

The lack of significance regarding HNR suggests that voice quality does not degrade, despite previous literature reporting reduced HNR as a hallmark of hypokinetic dysarthria in PD (Darley et al., 1969). Sustained vowel phonation may not be sensitive enough to capture voice quality changes.

Limitations

This study, while brief, is not wholly robust. Several limitations pervade this design. Admittedly, the dataset obtained for this analysis did not contain information about PwPD's severity or progression, making it difficult to understand deviations from prior findings in the literature. The dataset was also recorded using participants' phones, leading to more variability in recording quality. For example, participants could move the phone while recording and cause confounding changes in the sound. Capturing dynamic voice control was also limited, as participants only produced a sustained vowel. Further research should examine conversational speech for harmonicity.

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Conclusion

Despite previous literature suggesting that PwPD have reduced harmonicity and greater F0 and acoustic instability, no individual significant differences were detected after complete statistical analyses. Pitch control, as reflected by mean F0, may serve as a biomarker for laryngeal stiffness and rigidity in PD, yet further research is needed to verify this. MANOVA did reveal group differences not readily apparent in univariate tests, indicating that this analysis should be used for complex voice datasets in the future. Future research should aim to examine the variance within PD more closely while maintaining ecological validity with conversational speech.

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